

Divisibility & Digital Roots — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (b) the sum of its digits is divisible by 9.

Q2 (c) the alternating sum of digits is 0 or a multiple of 11.

Q3 1. $7+3+0+9 = 19 \rightarrow 1+9 = 10 \rightarrow 1$.

Q4 (c) 9. $4 + 0 + 5 = 9$.

Q5 True. $6 = 2 \times 3$, so both tests must pass.

Q6 (d) A false, R true. 1 has only one factor, so it is not prime; the reason (primes have two factors) is correct.

Short answer

Q7 (i) Yes (digit sum 9). (ii) No (digit sum 32, not a multiple of 9).

Q8 Smallest 1008, greatest 9999. (9×112 and 9×1111 .)

Q9 $z = 0$ or 9. Digit sum $9 + z$ must be a multiple of 9.

Long answer

Q10 Divisible by 3 only. Digit sum 30 $\rightarrow$ divisible by 3, not 9; ends in 9 $\rightarrow$ not 2, 4, 5, 6, 10; alternating sum 4 $\rightarrow$ not 11.

Q11 Digital root = 2. $63p$ and $18q$ are multiples of 9 (so contribute 0); 29 leaves remainder 2 on division by 9, giving digital root 2 whatever p and q are.

Q12 Smallest 100001, greatest 999999. (11×9091 and 11×90909 .)

Total: 21 marks.